

Fast Expense Tracker - Project Workflow

This document summarizes the frontend, backend, and database workflow for the Fast Expense Tracker project.

1. Frontend Workflow

The React frontend manages user authentication and expense interactions with the backend API.

- Axios client uses base URL: `http://localhost:8000`
- Request interceptor adds Authorization header with Bearer token from `localStorage`.
- Signup form calls `POST /auth/signup` and stores the returned token.
- Login form calls `POST /auth/login` and stores the returned token.
- Dashboard loads expenses via `GET /expense/` and performs create/delete actions.

2. Backend Workflow

FastAPI handles authentication, expense endpoints, and user management.

- FastAPI app includes auth, user, and expense routers.
- CORS allows requests from `http://localhost:5173`.
- Auth module hashes passwords and issues JWT bearer tokens.
- `get_current_user` uses `HTTPBearer` to validate Authorization header.
- Protected routes require a valid JWT and use `user_id` from token payload.
- `POST /auth/signup` creates a user and returns `access_token` and `token_type`.
- `POST /auth/login` validates credentials and returns `access_token` and `token_type`.
- Expense endpoints create, read, update, and delete expenses for the authenticated user.

3. Database Workflow

SQLAlchemy models and async database sessions persist users and expenses to PostgreSQL.

- User model stores id, email, and hashed password.
- Expense model stores id, title, amount, `created_at`, and `user_id`.
- `Expense.user_id` is a foreign key linking expenses to users.
- Startup event ensures tables are created using `Base.metadata.create_all()`.

4. Full Request Lifecycle

- User opens frontend and signs up or logs in.
- Frontend sends credentials to the backend auth endpoint.
- Backend validates and returns JWT access token.
- Frontend stores token and sends it on future API requests.
- Backend decodes token, extracts `user_id`, and authorizes the request.
- Backend reads/writes expense records linked to the authenticated user.
- Database persists user and expense data in separate tables.

5. Notes

Expenses are associated with the currently logged-in user by `user_id`, not by email directly. The frontend dashboard requests and displays only the authenticated user's expenses.